Mathematical Gems from OT4ML

75 possible blog posts

This catalogue collects all 75 ideas proposed after screening *Optimal Transport for Machine Learners*, in the spirit of the mathematical posts on Gabriel Peyré's blog. The editorial filter is simple: *one surprising statement, a proof with an identifiable mechanism, and a picture that reveals something the formula conceals*.

These are post proposals, not completed articles or claims of novelty. Some are classical results with an underexposed interpretation; others are counterexamples or synthesis pieces that deserve a more substantial mathematical note. The hypotheses and attributions in the linked book passages should be retained when a proposal becomes a post.

Where to Start

A first sequence: 1, 6, 10, 12 and 61. These combine concise proofs, distinctive mechanisms and straightforward visual or interactive illustrations: inversion penalties, lattice effects in the CLT, maximal correlation in Sinkhorn, failure of Hellinger displacement convexity, and the logarithmic mean in Markov diffusion.

Deeper mathematical notes: 2, 3 and 4. Scalar moment closure, non-Gaussian GW optimizers and the topology of a measure-level Min-SW extension are particularly promising. Their exact scope and literature positioning deserve extra care before publication.

Avoiding repetition. The list favors new angles over direct repetitions of existing posts on Gaussian preservation, Gaussian KL barycenters, Perron–Frobenius/Sinkhorn, moment measures, complete positivity, inverse gap losses and Muon. Related proposals are intended as sequels, contrasts or new connections.

How to use the pointers. Each entry has a printable theorem or section locator and a clickable link into the online book PDF. Numbering and printed page numbers refer to the current manuscript snapshot; the source is standalone and does not depend on the book's LaTeX files. PDF bookmarks give direct access to all 75 proposals.

Priority Candidates

1. The Monge gap is a sum of penalties for inversions

On a uniform grid $x_i = x_1 + (i-1)\Delta$, let $t_i = T(x_i)$ and $\alpha_n = n^{-1}\sum_i \delta_{x_i}$. For quadratic cost,

$$\mathcal{M}_{\alpha_n}^{|\cdot|^2}(T) = \frac{2\Delta}{n} \sum_{1 \le i < j \le n} (t_i - t_j)_+.$$

A comparison with optimal transport becomes a convex penalty for being out of order. Bubble sort explains the identity. The post can distinguish rearrangement (which preserves the output law), isotonic regression (which pools values), and Monge-gap-penalized regression.
Possible illustration. Reuse the kernel-regression regularization paths, with inversions highlighted.
Book pointer: Proposition 3.44, p. 62.

2. Which statistics have an autonomous Wasserstein evolution?

For $f(\alpha) = g(\int \varphi \, d\alpha)$, the evolution of the scalar moment closes for every smooth g and every admissible initial law precisely when

$$|\nabla \varphi|^2 = a + b\varphi.$$

The necessity follows by comparing two-point mixtures with equal moments, which forces a Jensen functional equation. The condition connects moment closure to the eikonal equation. Signed distance to a curved hypersurface supplies a non-polynomial, locally valid example, as long as the support remains in a smooth tubular neighborhood.
Book pointer: Proposition 16.30, p. 426.

3. Gaussian marginals need not have a Gaussian GW optimizer

For the distortion $(|x - x'|^2 - |y - y'|^2)^2$, couple a one-dimensional standard Gaussian to a sufficiently high-dimensional standard Gaussian. A coupling that correlates the radii can beat every jointly Gaussian coupling. Its advantage comes from a mixed fourth moment that covariance optimization cannot detect. This sharply separates Gaussian-constrained GW from unrestricted GW; the dimensional and dissimilarity assumptions must remain explicit.
Possible illustration. Plot the Gaussian optimum and the radial competitor against dimension.
Book pointer: Proposition 13.34, p. 300; discuss the Gaussian-optimality conjecture of Delon, Desolneux and Salmona cited there.

4. An exact Gaussian formula can coexist with empirical inconsistency

For the representation-invariant measure-level Min-SW extension used in the book, Min-$\mathrm{SW}_2 = \mathcal{W}_2$ on nondegenerate Gaussians. Yet in dimension $d \ge 2$, empirical measures can converge to the standard Gaussian in $\mathcal{W}_2$ while their Min-SW discrepancy from that Gaussian stays bounded away from zero. The obstruction is transverse quantization inside projected fibers. This is a warning about extending an empirical construction to all measures, not an unqualified statement about every sliced-transport definition.
Book pointer: Propositions 11.35–11.36, p. 230.

5. Why a hard density cap can have no effect on optimal transport

On a convex Euclidean domain, if both endpoint densities are at most κ, the quadratic displacement interpolation also respects this cap. A hard density constraint therefore leaves the pure Benamou–Brenier optimum unchanged in this setting. A determinant-concavity argument explains the phenomenon. Obstacles, nonconvex bottlenecks or additional objectives are needed to produce genuinely different congestion dynamics.

Possible illustration. Contrast straight interpolation in a convex domain with transport through a narrow doorway.

Book pointer: Proposition 15.22, p. 364.

6. Two symmetric central limit theorems, two different Wasserstein rates

For standardized sums of symmetric Bernoulli variables, $\mathcal{W}_1(\alpha_n, \mathcal{N}(0,1)) \sim 1/(2\sqrt{n})$; for a continuous uniform input, the leading order is C/n. Symmetry removes the skewness contribution in both cases, but a lattice leaves a first-order sawtooth error in the cumulative function. This makes the difference between lattice and density asymptotics visible without changing the limiting Gaussian.

Possible illustration. Combine the existing rate plot with a magnified cumulative-error profile.

Book pointer: Propositions 10.3–10.4, p. 188.

7. Why KL belongs to two different divergence families

Density-ratio divergences and Bregman divergences arise from different constructions. Under the smoothness and nondegeneracy assumptions of the book, KL is essentially their common member, up to scaling and irrelevant affine terms. Comparing second variations forces the Fisher-information form; an integration then identifies the entropy. The result explains why algorithms for the two divergence families coincide for KL and generally differ otherwise.

Book pointer: Proposition 8.34, p. 155.

8. Why exact alternating maximization can stop immediately

Hard c-transforms obey a triple-transform identity. Alternating exact best responses can consequently reach a closed pair without solving the full Kantorovich dual. The useful lesson is that coordinatewise optimality need not give global optimality for a nonsmooth concave problem. Entropic smoothing breaks this locking mechanism and allows repeated updates to keep improving.

Possible illustration. A small discrete dual with a coordinatewise fixed pair and a better joint direction.

Book pointer: Proposition 5.14, p. 94.

9. A nonconvex distance kernel produces a geodesically convex MMD

On the line, take the conditionally positive kernel $k(x, y) = -|x - y|$ with the energy-distance convention of the book. The self-interaction is affine in quantile coordinates, whereas the target cross-term is convex. The resulting squared MMD is displacement convex even though the kernel is not jointly convex. Ordering on the line, rather than ordinary kernel convexity, is the decisive mechanism.

Book pointer: Proposition 15.18, p. 360.

10. Sinkhorn slows down when one variable almost determines the other

The exact local factor for a full Sinkhorn cycle is the square of the maximal correlation of its optimal coupling. Conditional expectation and the law of total variance turn an algorithmic rate into a probabilistic statement: nearly deterministic transport leaves little conditional randomness, hence slow local convergence. This is a natural sequel to a global Hilbert-metric analysis rather than a repetition of it.

Possible illustration. Compare increasingly concentrated couplings and their measured local rates.

Book pointer: Proposition 9.20, p. 177.

11. Conditional expectation preserves quadratic transport optimality

If (X, Y) is an optimal coupling for squared Euclidean cost, the map $x \mapsto \mathbb{E}[Y \mid X = x]$ is optimal from the source to its own output law, even when the source has atoms. A variance decomposition proves the result. The qualification about the output law matters: it is generally not the original target, and the conclusion fails for arbitrary nonoptimal input couplings.

Possible illustration. Collapse each fan of transport segments onto its conditional barycenter.

Book pointer: Proposition 12.33, p. 273.

12. A Gaussian target does not make every divergence displacement convex

Let $\alpha_m = \mathcal{N}(m, 1)$ and $\beta = \mathcal{N}(0, 1)$. This translation family is a Wasserstein geodesic, but squared Hellinger divergence satisfies

$$H^2(\alpha_m, \beta) = 2(1 - e^{-m^2/8}), \qquad \mathrm{KL}(\alpha_m \mid \beta) = m^2/2.$$

The Hellinger curvature becomes negative for $|m| > 2$, in contrast to KL. Log-concavity of the target alone cannot transfer KL's displacement-convexity properties to every ϕ-divergence.

Possible illustration. The two scalar functions and their second derivatives.

Book pointer: Remark 15.21, p. 363.

Further Candidates

13. Composing optimal maps can destroy optimality

Two symmetric positive-definite linear maps are individually Brenier maps, but their product need not be symmetric. Noncommuting maps can introduce rotation through an intermediate measure. This gives an elementary matrix counterexample to the tempting idea that composing two optimal transport steps is still optimal. On the line, by contrast, increasing maps remain increasing under composition.

Book pointer: Remark 2.42 and the following example, p. 30.

14. Polar factorization: sorting a vector field

Under the appropriate nondegeneracy assumptions, a map factors into a measure-preserving rearrangement followed by the gradient of a convex function. The rearrangement changes labels without changing the distribution; the gradient part carries the distributional change. This is a higher-dimensional version of sorting, with a useful analogy to the polar decomposition of a matrix.

Possible illustration. The original vector field, its relabeling, and its monotone factor.

Book pointer: Proposition 2.28, p. 22.

15. Triangular rearrangement emerges from an extremely anisotropic metric

Knothe–Rosenblatt transport is obtained by matching successive conditional distributions. Under suitable assumptions, it also arises as the limit of quadratic optimal transport when the coordinate costs have increasingly separated scales. A singular perturbation of the geometry thus encodes an order of conditioning.

Possible illustration. Animate the transport as coordinate anisotropy increases toward a triangular map.

Book pointer: Proposition 2.52, p. 35.

16. The Gaussian Wasserstein formula is not specifically Gaussian

The same affine optimal maps and Bures covariance formula apply to elliptical laws sharing a common radial generator. Gaussianity is sufficient, not necessary. The shared-generator hypothesis is essential: covariance alone cannot determine quadratic transport between arbitrary elliptical laws.

Possible illustration. Compare transport between Gaussian ellipses and between uniform ellipsoids with the same covariance pairs.

Book pointer: Remark 2.57, p. 38.

17. Bures distance is a Procrustes problem between matrix square roots

Writing a covariance as $AA^\top$ leaves an orthogonal ambiguity in its factor. Minimizing the Frobenius distance between appropriately aligned factors gives the Bures distance. The same formula links Gaussian transport, shape alignment and quotient geometry. It is particularly instructive to contrast a fixed square-root difference with the optimally rotated one.

Book pointer: Proposition 2.60, p. 40.

18. Is a singular covariance nearby or infinitely far away?

A rank-deficient covariance is at finite Bures distance but lies infinitely far away in the Gaussian Fisher–Rao geometry. Already in one dimension, standard deviation gives the Wasserstein coordinate while its logarithm gives the Fisher–Rao scale coordinate. The same boundary therefore has very different metric meanings.

Possible illustration. Compare approaches to the boundary of the two-dimensional PSD cone.

Book pointer: Section 2.7, comparison with Fisher–Rao, p. 36.

19. Computing quadratic Wasserstein distance without inverting CDFs

The Bobkov–Ledoux formula expresses one-dimensional squared Wasserstein distance as a double integral involving positive parts of cumulative-function differences. A layer-cake argument turns the area associated with squared displacements into a CDF-only expression. This is an unexpectedly geometric alternative to the familiar quantile formula.

Possible illustration. Animate the regions counted by the two integral formulas.

Book pointer: Proposition 2.45, p. 31.

20. On a tree, optimal transport is bookkeeping across cuts

For the path metric on a weighted tree, $\mathcal{W}_1$ is the sum over edges of the edge length times the absolute mass imbalance on one side of the cut. Every unit of imbalance must cross that edge, and the resulting lower bound can be attained simultaneously. The formula gives a linear-time algorithm and explains why higher powers of the path distance require a different argument.

Book pointer: Proposition 2.50, p. 33.

21. Removing crossings still leaves exponentially many matchings

For alternating source and target vertices of a convex polygon, the noncrossing perfect matchings are counted by Catalan numbers. Eliminating proper crossings is therefore a powerful geometric simplification but not an efficient characterization of the optimum: exponentially many candidates can remain. A short counting argument connects planar transport to classical combinatorics.
Book pointer: Proposition 1.5 and the following Catalan discussion, p. 7.

22. Concave transport costs reward unequal travel distances

For concave powers of travel distance, one long trip and one short trip can cost less than two intermediate trips. The usual intuition behind monotone quadratic matching is reversed. Simple one-dimensional examples expose nested matching patterns and show why changing the exponent changes the combinatorics, not just the numerical values.
Book pointer: Section 1.1, concave costs on the line, p. 1.

23. Independence is almost never optimal transport

For a continuous cost on compact supports, the independent coupling is optimal only in the degenerate case where the cost is additive, $c(x, y) = u(x) + v(y)$ on the product support. In that case every coupling is optimal. Four-point exchanges prove this rigidity and explain why genuine transport must create dependence.
Book pointer: Proposition 3.29, p. 56.

24. Optimal transport is the largest convex extension of a pointwise cost

Prescribe the value $c(x, y)$ on pairs of Dirac masses. Among jointly convex, lower-semicontinuous extensions with the required Dirac bound, the Kantorovich value is pointwise largest under the hypotheses stated in the book. Jensen's inequality applied to a coupling is the key mechanism. This gives an axiomatic reason for the relaxation, beyond the usual optimization derivation.
Book pointer: Theorem 3.35, p. 59.

25. Transport potentials are always convex in a tropical sense

Finite pointwise minima of shifted c-concave functions remain c-concave. This stability holds even when ordinary convex combinations leave the class. The proof passes directly through the infimum defining the c-transform. It offers an accessible bridge between optimal transport, min-plus algebra and architectures built by taking minima of elementary potentials.
Book pointer: Proposition 5.17, p. 95.

26. Convexity of a family of functions can encode curvature of a space

Whether ordinary averages of c-concave potentials remain c-concave is tied to curvature conditions on the cost. Squared Euclidean distance is the simplest example; spheres and products reveal a less obvious geometric structure. An expository post can introduce the phenomenon without reproducing the full regularity theory, using the precise results and assumptions of Kim–McCann and Figalli–Kim–McCann cited in the book.
Book pointer: Section 5.3, paragraph on c-concave functions, p. 91.

27. Bernstein polynomials on the space of probability measures

Given a functional f, form $(B_n f)(\alpha) = \mathbb{E}[f(\hat{\alpha}_n)]$ from an empirical measure with n independent samples. This is a polynomial functional of α. Empirical Wasserstein bounds immediately give approximation rates for Wasserstein–Hölder functionals. The analogy with classical Bernstein polynomials is exact at the level of sampling and averaging, while the dimension-dependent rate exposes the difficulty of approximation on measure space.
Book pointer: Proposition 4.36 and its preceding definitions, p. 81.

28. Brenier maps form a closed convex cone

Fix an absolutely continuous source. Convex combinations of its square-integrable Brenier maps are still gradients of convex functions and are optimal to their own pushforward laws. The range of linear optimal-transport coordinates is closed and convex in the reference L^2 space. This rigorously justifies map interpolation and averaging, without identifying these operations with arbitrary Wasserstein geodesics or barycenters.
Book pointer: Proposition 11.42, p. 235.

29. Weak changes in a law can control strong changes in its transport map

Under the source-density and compact convex-domain assumptions stated in the book, a $\mathcal{W}_1^{1/4}$ bound controls the L^2 change in the associated Brenier maps. Stability of the optimal value is much easier than stability of the optimizer; this theorem bridges the two. A useful post would explain what the exponent measures and why the reference source must be controlled.
Book pointer: Proposition 11.43, p. 236, with the survey and sharper stability result cited there.

30. Three Hilbert embeddings, three geometries of distributions

MMD averages kernel features, sliced Wasserstein records projected quantiles, and linear OT records reference-to-target maps. All three admit Hilbert-space descriptions, but they preserve different information and have different image sets. Consequently, averaging coordinates has a different meaning in each representation. The comparison makes clear that "Hilbertian" is not itself a complete geometric description.
Book pointer: Remark 11.44, p. 237.

31. Particles and smooth densities have different local sliced geometries

Along suitable finite atomic curves with fixed weights, the sliced metric derivative equals the Wasserstein metric derivative divided by $\sqrt{d}$. For smooth anisotropic Gaussian deformations, the corresponding inequality can be strict. Projected ordering explains the atomic equality, while averaging conditional velocities explains the loss for densities. The contrast warns against inferring continuum tangent geometry from finite particles alone.
Book pointer: Remarks 11.28–11.29, p. 225.

32. Averaging projected distributions need not produce a distribution

Independently averaging quantiles in each projection direction need not produce a Radon-consistent family. Reconstructing the result by a pseudoinverse may even leave the cone of nonnegative densities. The surprise is that individually valid one-dimensional distributions need not fit together into any multidimensional law. This connects sliced barycenters to tomography and compatibility constraints.
Book pointer: Proposition 12.15 and its surrounding Radon discussion, p. 255.

33. Optimal quantization uses averages of quantiles, not midpoint quantiles

For n equally weighted atoms on the line, the optimal location of atom j is $n \int_{(j-1)/n}^{j/n} C_\alpha^{-1}(r)\, dr$, rather than the midpoint quantile in general. The proof is an orthogonal projection of the quantile function onto piecewise constant functions. This gives a precise distinction between optimal deterministic quantization and a common sampling heuristic.
Book pointer: Proposition 6.14, p. 110.

34. Carefully placed particles beat random particles by a full power

For regular one-dimensional laws satisfying the relevant quantile assumptions, optimal equal-weight quantization has squared Wasserstein error of order n^{-2}, while the expected squared error of independent particles is of order n^{-1}. Quantile approximation gives the first rate; a Brownian-bridge limit gives the second. The statement should retain the regularity and integrability conditions, rather than suggesting a universal rate for all one-dimensional laws.
Book pointer: Propositions 6.14–6.15, p. 110.

35. A discrete barycenter can be much sparser than its candidate support

For S discrete input laws with support sizes n_s, a barycenter exists with at most $\sum_s n_s - S + 1$ atoms. This can be dramatically smaller than the product-sized candidate set suggested by all tuples of input points. The proof is a rank count for a multimarginal linear program, connecting transport sparsity to basic feasible solutions.
Book pointer: Corollary 12.22, p. 260.

36. Multimarginal transport meets graphical models

When a multimarginal cost factors along a graph, the Gibbs tensor also factors. Message passing can then compute Sinkhorn contractions without constructing the full tensor. The decisive exponential dependence is governed by treewidth, not necessarily by the number of marginals. Trees, cycles and dense graphs make the contrast concrete and connect transport to graphical models and tensor networks.
Book pointer: Section 12.2 and Example 12.24, p. 257.

37. Sinkhorn divergences forget the reference measure

Changing a compatible product reference in entropic OT adds marginal KL terms to the value. These terms cancel exactly when the two self-costs are subtracted to form the Sinkhorn divergence. The result explains why apparently different entropy conventions can yield the same debiased discrepancy, provided the absolute-continuity and finiteness assumptions are satisfied.
Book pointer: Remark 8.38, p. 159.

38. An entropic transport problem requiring no Sinkhorn iterations

Choose a Gibbs transition from the source, normalize it appropriately, and take its output law as the target. The resulting coupling already solves the entropic transport problem. For quadratic cost and a Lebesgue reference, this includes Gaussian noising. The example links a closed-form Schrödinger scaling to the noising paths used in diffusion models.
Book pointer: Proposition 8.14, p. 138.

39. Large-temperature Sinkhorn projects onto interaction terms

The first correction to the independent coupling is governed by the cost after subtracting its source average and target average and restoring its global average. This double centering is the interaction term of an ANOVA decomposition. A large-temperature expansion thus has a simple Hilbert-space interpretation: row-only and column-only information cannot change the coupling.
Book pointer: Proposition 8.16, p. 140.

40. Fisher information hides in the small-noise transport bias

For sufficiently regular quadratic transport, the second-order entropic correction, after the appropriate endpoint and normalization terms are removed, involves Fisher information integrated along the displacement interpolation. The bias therefore detects the roughness of intermediate densities, not just the endpoint displacement. A careful post should state the normalization before displaying the expansion: it determines which terms appear at each order.
Book pointer: Proposition 8.17, p. 141.

41. Why half the temperature can give a better barycenter

In a doubly regularized barycenter, an outer entropy with strength $\tau = \epsilon/2$ cancels the first-order candidate-entropy contribution of the inner entropic transport costs, under the small-temperature expansion's hypotheses. A seemingly arbitrary parameter choice thus has an asymptotic explanation. Contrast sharpening, entropy cancellation and uniqueness, which are related but distinct issues.

Book pointer: Definition 12.13 and the subsequent discussion, p. 252.

42. Eliminating a Sinkhorn potential improves conditioning computably

The semi-dual Hessian at the optimum is $(I - T^*T)/\epsilon$, where T is conditional expectation under the optimal coupling. Variable projection turns the full Hessian into a Schur complement. Because its diagonal blocks are identities after normalization, the spectral conditioning bounds can be compared explicitly. Distinguish a sharp comparison of available bounds from an unconditional equality between actual condition-number improvement factors.

Book pointer: Proposition 9.24, p. 181.

43. Sinkhorn acceleration is classical SOR in disguise

Over-relaxing each of Sinkhorn's two block updates gives a nonlinear analogue of successive over-relaxation. Linearization explains the optimal parameter and the square-root improvement in a small local spectral gap. This provides a clean reason for acceleration without introducing an unrelated momentum model. The local nature of the optimal-rate statement should remain explicit.

Possible illustration. Reuse the marginal-error curves together with predicted and measured gains.

Book pointer: Proposition 9.22, p. 179.

44. What happens when the Sinkhorn temperature becomes complex?

Near a positive real temperature, gauge-fixed Sinkhorn scalings have a local holomorphic continuation under the hypotheses of the continuation theorem. The implicit-function mechanism is familiar, but its application to couplings produces striking oscillatory images. The continuation is local: it is not a license to substitute any complex temperature into a real formula or to assume global convergence of complex Sinkhorn iterations.

Book pointer: Theorem 8.43, p. 161, with the Carlier–Pegon–Tamanini attribution given there.

45. Allowing mass variation can make Sinkhorn convergence simpler

KL penalties on the marginals introduce damping factors into unbalanced Sinkhorn updates. They break the balanced problem's additive gauge and yield a direct contraction in the potential sup norm in the stated bounded setting. The result contrasts with the projective geometry needed for balanced scaling: relaxing the constraints can simplify the convergence mechanism.

Book pointer: Proposition 11.12, p. 217.

46. Changing the regularizer changes whether transport has exact zeros

Entropy produces an exponential primal–dual law and hence dense couplings under finite-cost positivity assumptions. A quadratic regularizer instead produces thresholding and can create exact zeros. Generalized soft c-transforms connect this distinction to one-dimensional monotone equations and simplex projections. Sparsity here is an analytic consequence of the regularizer, not merely a numerical threshold.

Book pointer: Section 8.8 and Proposition 8.30, p. 153.

47. Unbalanced transport separates shape from a scalar mass problem

With equal KL marginal penalties of strength τ, write the input masses as A and B and optimize first over a normalized coupling shape. If its optimal shape cost is E, the transported mass is $M^* = \sqrt{AB}\exp(-E/(2\tau))$ in the book's normalization. Geometry enters a one-dimensional mass minimization as an exponential discount. The geometric mean of the input masses is the unpenalized mass scale.

Book pointer: Proposition 11.3, p. 210.

48. The Hellinger interpolation of two Gaussians has three components

Hellinger interpolation is linear in square-root densities. Squaring a combination of two Gaussian square roots gives the two endpoint densities and a Gaussian geometric-mean cross-term: an unnormalized three-Gaussian mixture. This is a transparent counterexample to Gaussian closure for pure Fisher–Rao/Hellinger motion. Its relation to WFR should be phrased as a limiting obstruction, not as an explicit formula for every WFR geodesic.

Book pointer: Remark 11.18, p. 219.

49. Partial transport trades a price against transported mass

The optimal cost at prescribed transported mass is a convex value function. Adding a linear price for unmatched mass selects a supporting slope of this curve. Trimming arguments connect this formulation to TV-penalized unbalanced transport under the assumptions of the equivalence result. The scalar value curve makes clear when several transported masses can share the same price.

Possible illustration. Animate the active transport region as the price changes.

Book pointer: Proposition 11.21, p. 220.

50. Forgetting mixture labels is a Wasserstein contraction

A probability law over probability laws can be collapsed to its mixture. Couple the outer laws and then glue suitable inner couplings: the resulting coupling proves that collapse is non-expansive for Wasserstein-over-Wasserstein. The proof identifies precisely what is forgotten when hierarchical labels are discarded. It is also a compact example of gluing as the engine of a metric comparison.

Book pointer: Proposition 13.10, p. 286.

51. Compare shapes through distributions of distance distributions

Associate to each point the law of its distance to a random point. A shape then becomes a distribution over one-dimensional distributions. Mémoli's profile bound compares these objects and lower-bounds intrinsic GW. Conditional couplings explain the inequality, while the construction connects directly to Wasserstein-over-Wasserstein.

Possible illustration. Reuse the cat–bunny figure with matched points and colored distance profiles.

Book pointer: Proposition 13.24, p. 292.

52. A nonnegative distortion can define a concave coupling objective

Squared GW distortion is pointwise nonnegative, yet its quadratic form can be concave on zero-marginal directions when the dissimilarities have the required conditional definiteness. A tensor-feature calculation turns the second variation into a negative squared norm. This conditional concavity explains extreme-point optimizers and permutation solutions in appropriate finite uniform settings, without making the optimization convex or globally easy.

Book pointer: Proposition 13.27, p. 295.

53. Euclidean GW can be understood as learning a cost matrix

For inner-product or squared-distance dissimilarities, completing a square reduces the learned-cost representation of GW to a finite-dimensional matrix parameter with a quadratic penalty. The remaining OT problem uses a cost depending on that matrix. This interpretation separates cost learning from coupling computation and reveals why these Euclidean models are unusually tractable, although the outer optimization is still nonconvex.
Book pointer: Proposition 13.32, p. 298.

54. Duplicating a variable can exactly relax a nonconvex problem

Replacing the diagonal quadratic GW objective by a two-coupling bilinear objective gives a lower bound and a problem convex in either block separately. Konno's polarization argument shows tightness in the conditionally concave regime. With positive entropy, strict convexity additionally constrains minimizing block pairs. The distinction between exactness of the relaxation and global convergence of alternating minimization is an essential part of the story.
Book pointer: Proposition 13.39, p. 303.

55. Weak transport measures distance to a mean-preserving spread

Quadratic barycentric weak transport equals the minimum of $\mathcal{W}_2(\alpha, \eta)^2$ over $\eta \preceq_{\mathrm{cx}} \beta$. It vanishes exactly when a martingale coupling exists, subject to the standard moment assumptions. Conditional Jensen gives one direction; the convex-order characterization supplies the other. The result connects a convex transport relaxation to mean-preserving spreads and mathematical finance.
Book pointer: Proposition 12.44, p. 278.

56. The right topology does not guarantee the triangle inequality

A quantum-inspired classical transport cost compares square-root coupling densities with their transposes. It is convex in the coupling, nonnegative and separating; Wasserstein comparisons give the desired topology in the stated setting. Nevertheless its square-root version fails the triangle inequality already on three points. The example separates topological usefulness from metric structure particularly cleanly.
Book pointer: Propositions 13.53 and 13.56, Example 13.57, p. 310.

57. Replacing scalar scalings by matrices changes Sinkhorn's meaning

Classical row and column scalings commute in ways that matrix-valued updates do not. In quantum OT, operator scaling is generally not the same as exact Bregman projection for the entropically regularized transport objective. The diagonal case hides this distinction because it reduces to classical Sinkhorn. A two-by-two noncommuting example could explain why copying a scalar formula does not automatically copy its variational interpretation.
Book pointer: Section 13.4, Gurvits scaling and quantum Sinkhorn, p. 308.

58. Particles can move while their probability law stands still

A rotating cloud can have a stationary radial distribution. Its particle velocity is therefore not the intrinsic velocity of the measure curve. The minimum-kinetic-energy representative removes invisible circulation and is a gradient, characterized by a weighted Poisson problem. This is a tangible explanation of the quotient hidden in Wasserstein tangent geometry.
Book pointer: Proposition 14.3, p. 322.

59. The same perspective transform convexifies transport and divergences

The momentum action $|m|^2/\rho$ and the divergence integrand $q\phi(p/q)$ are instances of the same perspective construction. Its convexity explains both the Benamou–Brenier momentum formulation and joint convexity of ϕ-divergences. Concave mobilities and unbalanced actions extend the same mechanism. A single geometric proof can replace several apparently unrelated convexity calculations.

Book pointer: Proposition 14.13 and its cross-reference to divergence convexity, p. 332.

60. Why random-path optimization reduces to endpoint optimization

The KL chain rule splits path-space entropy into endpoint entropy and the average conditional-bridge entropy. At fixed endpoints, the optimum keeps the reference bridges unchanged. The Schrödinger problem consequently reduces to an entropic coupling problem, followed by a bridge lift. This gives a clean reason why a finite-dimensional-looking Sinkhorn problem can describe a law on an infinite-dimensional path space.

Book pointer: Proposition 14.11, p. 329.

61. The logarithmic mean is forced by linear Markov diffusion

To make entropy descent produce a linear jump flux, the edge mobility must satisfy $\theta(a, b)(\log a - \log b) = a - b$. This identifies the logarithmic mean, including its continuous extension on the diagonal. A seemingly specialized mean is therefore selected by a chain rule. The same identity connects discrete Markov chains and continuum jump kernels.

Possible illustration. An edge with unequal densities, its entropy gradient, and the resulting linear flux.

Book pointer: Proposition 15.58 and the following Markov-chain discussion, p. 388.

62. Fractional heat is entropy descent in a different transport geometry

Replace local velocities by antisymmetric jump fluxes and use a reversible power-law jump kernel. Entropy descent then produces a fractional diffusion operator rather than the ordinary Laplacian. The energy can remain familiar while the transport geometry changes. An accessible post can derive the flux identity before discussing the fractional heat equation.

Book pointer: Section 15.6, p. 388.

63. One heat equation, two gradient-flow structures

The heat equation is both L^2 descent of Dirichlet energy and Wasserstein descent of entropy. A gradient flow is therefore specified by an energy and a geometry together, not by the PDE alone. Deriving the same equation twice gives a short and useful introduction to metric-dependent steepest descent.

Possible illustration. Compare what a unit tangent displacement means in the two geometries.

Book pointer: Remark 15.14, p. 354.

64. A spectral gap is a linearized entropy inequality

Perturb a minimizer of a Wasserstein–PL inequality and retain the quadratic terms. For relative entropy, logarithmic Sobolev linearizes to Poincaré, which is a spectral gap for the target-weighted Laplacian. The mechanism connects nonlinear entropy decay, local quadratic forms and linear operator theory. This can serve as a sequel to a PL post rather than another abstract convergence theorem.

Book pointer: Proposition 15.44, p. 376 and Proposition 15.48, p. 377.

65. A convex functional of measures can have bad particle equilibria

The functional $f(\alpha) = \int V \, d\alpha$ is affine in the measure even if V has nonglobal local minima. Particles can become trapped at those minima. Convexity under mixing measures is not convexity under moving their support. This elementary example explains why affine measure convexity alone is not a guarantee of global convergence for particle training.

Book pointer: Proposition 15.51 and its surrounding discussion, p. 381.

66. Mean–covariance closure does not require Gaussianity

If $f(\alpha) = g(m_\alpha, \Sigma_\alpha)$, its Wasserstein descent velocity is affine in space for every input law. Mean and covariance therefore obey an autonomous system even for non-Gaussian initialization. Gaussianity is only what turns these two statistics into a complete description of the law; other initial shapes undergo an affine deformation. This cleanly separates moment closure from distributional closure.

Book pointer: Equation (16.43), distribution-free moment closure, p. 425.

67. Smooth random paths and Brownian paths can share all time marginals

A static Gaussian noise variable reused along an interpolant can produce the same one-time laws as an Ornstein–Uhlenbeck noising process, despite completely different temporal dependence and quadratic variation. Matching marginals does not identify a path-space law. The example is an effective antidote to confusing a prescribed distributional path with a unique stochastic process.

Book pointer: Proposition 16.9, p. 405.

68. Gaussian flow matching detects noncommuting covariances

For the independent Gaussian interpolation and scalar schedules studied in the book, the resulting terminal transport map is Brenier precisely under the stated covariance commutation condition. Noncommutativity introduces a geometric obstruction that changing the scalar schedule cannot generally remove. This is a sharper test of transport optimality than simply checking that the endpoint marginals are correct.

Book pointer: Proposition 16.7, p. 404.

69. Gaussian blurring mean shift eventually makes one cluster

For blurring mean shift with a positive Gaussian kernel and compactly supported initial data, the invariant convex hull gives a uniform overlap bound, and a Dobrushin contraction yields exponential consensus. Several visible modes can therefore be long transients rather than permanent clusters. This contrasts with fixed-data mean shift and with bounded-confidence kernels, for which the same global consensus conclusion need not hold.

Book pointer: Remark 16.19, p. 416.

70. Unbiased samples do not give unbiased transport gradients

An empirical measure inserted inside a nonlinear transport optimization does not generally give an unbiased estimate of the population gradient. This is different from the usual stochastic-gradient story for an average of losses. A multimodal example can show how minibatches alter both the coupling and the induced update, rather than adding only centered noise to a fixed gradient.

Book pointer: Section 16.2, one-step models and minibatch transport, p. 407.

Longer Posts and Follow-Ups

71. The basic GW iteration is a difference-of-convex algorithm

In the conditionally concave regime, the coupling objective has a global affine upper model at each iterate. Minimizing that frozen OT model is a classical DC/majorization step; Toland duality supplies a complementary cost-side view. The connection clarifies descent and fixed-point conditions without confusing stationarity with global optimality. It also distinguishes this interpretation from alternating minimization of an arbitrary biconvex relaxation.
Book pointer: Remark 13.31, p. 297.

72. Optimal dual potentials determine statistical fluctuations

On finite supports, with unique normalized dual potentials, the two-sample limiting variance is

$$\mathrm{Var}_\alpha(f^*) + \mathrm{Var}_\beta(g^*)$$

for equal sample sizes and the book's normalization. Dual potentials act as influence functions. Nonuniqueness can instead produce non-Gaussian limits. The post should explain why fixed support sizes matter, rather than treating this CLT as a dimension-free theorem for arbitrary continuous transport.
Book pointer: Proposition 10.18, p. 200.

73. Gaussian attention, mean shift and denoising share one identity

For Gaussian kernel averaging with variance ϵ, the normalized local mean satisfies

$$m_\epsilon[\alpha](x) - x = \epsilon \nabla \log \rho_\epsilon[\alpha](x).$$

Differentiating the Gaussian convolution is enough to prove it. The same identity interprets a weighted average as a score step, connects attention to mean shift, and gives the denoising interpretation associated with Tweedie's formula. The kernel convention must be fixed to get the factor ϵ right.
Book pointer: Equation (16.41), attention and mean shift, p. 416.

74. Training a generator does not define an intrinsic distributional flow

Two parameterizations representing the same law can induce different particle and measure velocities under parameter gradient descent. The Eulerian velocity is a conditional expectation of parameter-induced particle motion. This formula pinpoints where the parameterization enters and explains why Euclidean training of a generator is not automatically Wasserstein steepest descent of its output law.
Book pointer: Proposition 16.12, p. 408.

75. An irrelevant additive constant becomes a growth rate

For probability measures, first variations are defined only modulo additive constants, because admissible mass variations have total mass zero. In an unbalanced transport geometry, the same constant affects reaction and therefore total mass. The passage from transport to transport–reaction turns a harmless gauge into a genuine dynamical parameter. A one-line energy shift gives a particularly clear illustration.
Book pointer: Section 15.7, p. 390.